Can fixed-input prediction-powered intervals be shortened? Headroom, finite-sample cost and validity

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Abstract

Prediction-powered inference (PPI, PPI++) gives valid confidence intervals from a small labelled set, a large unlabelled set and a fixed predictor. We ask whether, with all three held fixed and nominal coverage required, these intervals can be shortened, by how much and under what conditions. Viewing PPI++ as the best control variate that is affine in the predictor, we measure the headroom to the efficiency bound and the finite-sample price of learning a better control variate from the labels, with every width read as a paired ratio against PPI++ calibrated the same way and next to its coverage. For the mean on seven real pools the headroom, for the adjuster classes we tried, is at most about 1.2%, and on the four main pools cross-fitted adjusters are significantly wider than PPI++ in 252 of 270 cells for $n \leq 1000$ (never more than 0.55% narrower): there, no usable shortening exists. Where headroom exists, in populations with known truth, it is realised only above a threshold labelled size and by an adjuster matched to it: up to 26% narrower when the outcome is far from affine in the predictor, and 22% to 29% when features carry signal the predictor misses (from $n = 50$ with a linear adjuster), at coverage within 1.5 points of PPI++. For OLS coefficients on census data the gain is 2.8% at $n = 1000$. Under label shift PPI++ and the adjusters lose coverage. All results come from the released code.

1 Introduction

Prediction-powered inference (PPI) turns a small labelled sample and a large unlabelled one, scored by a fixed machine-learning predictor f, into a valid confidence interval that is shorter than the labelled-only interval [Angelopoulos et al., 2023a]; PPI++ tunes the weight of the predictor and is, among corrections that are affine in f, the shortest of them [Angelopoulos et al., 2023b]. We ask a question that sits after that: *with the labelled set, the unlabelled set and the predictor all held fixed, and nominal coverage required, can the PPI/PPI++ interval be made shorter, by how much, and under what conditions?* Nothing is bought here with more labels, a better f or a different labelling design; the only freedom is what the analyst does with the same records.

The answer we find is conditional, and for the most common use it is negative. Writing the estimator as a control variate (Sec. 3), PPI++ is the best affine function of f, and a known efficiency bound limits any control variate built from (f, X); the headroom is the gap between the two. On the seven real pools of the `ppi-python` package that gap, measured for the adjuster classes we tried (fitted on half-pools, so a lower bound on the true gap), is at most about 1.2% of the interval width for the mean (Sec. 5.2), and learning a better control variate from the n labels costs more than it returns: of 270 (pool, n, adjuster, calibration) cells on the four main pools at $n \leq 1000$, 252 are significantly *wider* than PPI++ calibrated the same way and none is more than 0.55% narrower (Sec. 5.3). Where the headroom is large, which we can only establish in populations with known truth, it is realised, but

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

only above a threshold n and only by an adjuster that matches its shape: up to 26% narrower when $E[Y|f]$ is far from affine, and 22% to 29% when features X carry signal that f misses, for n from 50 (a linear adjuster on (f, X)) to several hundred (flexible adjusters), at coverage within 1.5 points of PPI++'s (Sec. 5.4). On a regression problem with real data (OLS of income on covariates) the saving is 2.8% at $n = 1000$ (Sec. 5.5).

Contributions. (i) A variance calculus that places PPI++ inside a family of cross-fitted control variates, states the efficiency floor for the mean (a known bound, restated) and a first-order account of the estimation cost (Sec. 3). (ii) A like-for-like protocol: every width is a paired ratio against PPI++ with the same interval calibration (Wald, t or studentised bootstrap) and is read together with coverage and its Wilson interval, because a narrower interval that undercovers is not a saving (Secs. 4, 5.1). (iii) The measured headroom, finite-sample cost and break-even n on the real pools, in synthetic populations whose theory is known, and for OLS coefficients (Secs. 5.2–5.5). (iv) A statement of when the answer is "no" (calibrated or near-affine f, no informative X, small n, label shift) and every cell behind it, with the code that reproduces them. Positive and negative results are reported with equal weight; the claims are limited to the estimands, adjuster classes and sampling protocol described in Secs. 4–5 and in the discussion.

2 Related work

Prediction-powered inference. PPI [Angelopoulos et al., 2023a] combines a small labelled sample with predictions on a large unlabelled one to obtain valid intervals for means, quantiles and regression coefficients, whatever the predictor. PPI++ [Angelopoulos et al., 2023b] tunes the weight placed on the predictions so that the interval is never wider (asymptotically) than the labelled-only interval, and gives the efficiency bound and equality condition that frame our question. Variants change what is fixed: cross-prediction-powered inference [Zrnic and Candès, 2023] handles predictors trained on the same labelled data by cross-fitting; active statistical inference [Zrnic and Candès, 2024] chooses which records to label, and stratified PPI [Fisch et al., 2024] reweights strata. All of these either add information (a better predictor, a choice of labels) or change the sampling design. We keep the labelled set, the unlabelled set and the predictor fixed and ask only how the available information is *used*.

Semi-supervised and doubly-robust inference. Semi-supervised estimation of a mean from labelled and unlabelled covariates, with an efficiency bound, is studied by Zhang et al. [2019], and for linear regression by Chakrabortty and Cai [2018]. The cross-fitted, orthogonal-score construction behind our adjusters is that of double/debiased machine learning [Chernozhukov et al., 2018]. From this view PPI++ is the special case in which the nuisance function is restricted to be affine in f; the question is how much of the remaining distance to the bound can be recovered from the same labelled data, and at what finite-sample cost. We read the abstracts of the semi-supervised papers and use them for positioning only.

3 Setting and a variance calculus for fixed-input intervals

Setting. We observe n labelled pairs (X_i, Y_i) and N unlabelled covariates $\tilde{X}_j$, both i.i.d. from the same population (labels missing completely at random), and a fixed predictor f evaluated on every record, so every record carries $W = (f(X), X)$. The target is $\theta = \mathbb{E}[Y]$. Nothing is added: no labels, no unlabelled records, no change to f. Write $\rho = n/N$.

Control-variate form. For any fixed $g(W)$ and $\lambda \in \mathbb{R}$,

$$\hat{\theta}_{g,\lambda} = \bar{Y}_n - \lambda\big(\bar{g}_n - \tilde{g}_N\big), \tag{1}$$

where $\bar{g}_n$ and $\tilde{g}_N$ are the means of g on the labelled and unlabelled sets. It is unbiased for every λ. Classical inference is $\lambda = 0$; PPI [Angelopoulos et al., 2023a] is $g = f,\ \lambda = 1$; PPI++ [Angelopoulos et al., 2023b] is $g = f$ with λ tuned. Hence ***PPI++ is the best control variate affine in*** f. Its variance is

$$\mathrm{Var}(\hat{\theta}_{g,\lambda}) = \frac{1}{n}\Big[\mathrm{Var}(Y) - 2\lambda\,\mathrm{Cov}(Y, g) + \lambda^2(1+\rho)\,\mathrm{Var}(g)\Big], \tag{2}$$

minimised at $\lambda^\star = \mathrm{Cov}(Y,g)/\{(1+\rho)\mathrm{Var}(g)\}$, giving

$$\mathrm{Var}^\star(g) = \frac{\mathrm{Var}(Y)}{n}\Big(1 - \frac{R_g^2}{1+\rho}\Big), \qquad R_g^2 = \mathrm{corr}^2(Y, g(W)). \tag{3}$$

Proposition 1 (Asymptotic width ratio)**.** *For two control variates g_1, g_2 the ratio of asymptotic interval widths (same n, N, level) is $\sqrt{(1 - R_{g_1}^2/(1+\rho))/(1 - R_{g_2}^2/(1+\rho))}$.*

Proposition 2 (Floor)**.** *For every g, $R_g^2 \le \eta^2 := \mathrm{Var}(\mathbb{E}[Y \mid W])/\mathrm{Var}(Y)$, with equality iff g is an affine function of $\mathbb{E}[Y \mid W]$. Therefore* $\mathrm{Var}^\star(g) \ge \frac{1}{n}\{\mathrm{Var}(Y \mid W) + \frac{\rho}{1+\rho}\mathrm{Var}(\mathbb{E}[Y \mid W])\} = \mathrm{Var}(Y \mid W)/n + \mathrm{Var}(\mathbb{E}[Y \mid W])/(n+N)$.

The bound is the semiparametric efficiency bound for θ when labels are missing completely at random and the unlabelled records reveal only W (nonparametric model for (W, Y)). A bound of this form, with the equality condition that the control variate be proportional to $\mathbb{E}[Y \mid W]$, is given for general M-estimation targets by Angelopoulos et al. [2023b]; we restate it for the mean, as a standard result and not as a contribution, prove the control-variate form of the floor in Appendix A and verify it numerically in Sec. 5. What is new here is the *attainable* part: how close an adjuster learned from the fixed labelled set gets to the bound, at which n, and at what coverage. *Consequence.* The headroom over PPI++ is exactly $R_{\text{aff}}^2 \to \eta^2$, where $R_{\text{aff}}^2 = \mathrm{corr}^2(Y, f)$ if only f is used. It is zero if $\mathbb{E}[Y \mid W]$ is affine in f (e.g. a calibrated binary f with no informative X). Since $R^2/(1+\rho)$ is what enters (3), the attainable relative narrowing is largest for small ρ (few labels per unlabelled record) and fades as $\rho \to \infty$.

Estimating g costs width. A richer g must be learned from the same n labels. With K-fold cross-fitting, $\hat{g}^{(-k)}$ is fitted off fold k, so $\hat{\theta}$ stays unbiased. To first order the variance inflates by the excess risk of $\hat{g}$: $\mathrm{Var}(Y - \hat{g}) \approx \mathrm{Var}(Y - g^\star) + \mathbb{E}\|\hat{g} - g^\star\|^2$, so a gain over the affine tuning needs $(\eta^2 - R_{\text{aff}}^2)\mathrm{Var}(Y)$ to exceed an estimation term that grows with the adjuster's complexity and shrinks with n. This is a heuristic first-order argument, not a theorem: we use it only to interpret the measured break-even n (Sec. 5) and do not test it as a quantitative prediction.

Intervals. Wald intervals use the plug-in variance of (3); their coverage is asymptotic and degrades for skewed or rare-event Y and small n. We therefore also use a studentised bootstrap and always report width next to coverage: a shorter interval that undercovers is not a saving.

4 Method: cross-fitted control variates and calibrated intervals

The route we test uses nothing beyond the fixed inputs. It interprets "predictor fixed" as: f is not retrained and no label or unlabelled record is added, but a post-hoc map of $W = (f(X), X)$ fitted on the labelled set is allowed. Under the control-variate form of Sec. 3 this is a choice of g. If one insists on g affine in f, the result is PPI++ itself, which is therefore our baseline and the $K=5$ affine adjuster below is its ablation.

Cross-fitted adjusters. Split the labelled set into $K = 5$ folds. For each fold k fit an adjuster $\hat{g}^{(-k)}$ on the other folds, score the held-out labelled points with it, and score every unlabelled point with the fold average $\bar{g}_u = K^{-1}\sum_k \hat{g}^{(-k)}$. Writing m_i for the out-of-fold score of labelled point i, $\hat{\theta} = \bar{Y}_n - \hat{\lambda}(\bar{m}_n - \bar{g}_{u,N})$, with $\hat{\lambda}$ the plug-in version of $\lambda^\star$ in Sec. 3 and variance estimate $\widehat{\mathrm{Var}}(Y - \hat{\lambda}m)/n + \hat{\lambda}^2\widehat{\mathrm{Var}}(\bar{g}_u)/N$. Because $\hat{g}^{(-k)}$ never sees the points it scores on the labelled side, the estimator stays centred and the unlabelled mean is free. The adjuster classes (all fitted on the labelled set only) are: affine in f (`lin_f`); monotone, isotonic in f (`iso_f`); cubic polynomial in f (`poly_f`); gradient boosting on f (`gbm_f`); linear in (f, X) (`lin_fX`); gradient boosting on (f, X) (`gbm_fX`). Features X exist only for the census pools.

Interval calibration. Each estimator has three intervals: Wald ($z_{0.975}$), t with $n-1$ degrees of freedom, and a studentised bootstrap that resamples labelled pairs (Y_i, m_i), perturbs the unlabelled mean by its normal approximation and re-tunes λ in every resample. The same three calibrations are applied to PPI++ ($g = f$). Headline narrowing statements compare an adjuster with *PPI++*

Table 1: Pools used for the mean. Skewness and $\mathrm{corr}(Y, \hat{Y})$ are computed on the full pool; $\dim X$ is the number of features available to the adjusters.

pool	N_{pool}	outcome	mean Y	skew Y	$\mathrm{corr}(Y, \hat{Y})$	$\dim X$
galaxies	16,743	binary	0.259	1.1	0.74	0
forest	1,596	binary	0.152	1.9	0.54	0
alphafold	10,802	binary	0.177	1.7	0.82	0
gene_expression	61,150	continuous	6.77	0.56	0.96	0
census_income	380,091	continuous	4.31×10^4	4.7	0.61	2
census_healthcare	318,215	binary	0.671	-0.73	0.71	2
census_education	318,215	continuous	17.8	-1.8	0.63	4

calibrated the same way, so that a gain from calibrating the interval is not credited to the control variate; the published practitioner baseline (PPI++, Wald) is reported alongside and flagged wherever it undercovers.

OLS coefficients. For the secondary estimand the labelled OLS estimator is corrected by a vector of control variates $G_i = H^{-1} x_i (m_i - x_i^\top \hat{\beta})$, where H is the Gram matrix of all covariates (observed for unlabelled records too) and m_i imputes Y_i (by f_i, or by a cross-fitted adjuster). Each coordinate uses a variance-minimising combination of either its own control variate (the coordinate-wise PPI++ form) or all d of them. The implementation is checked against `ppi_py` 0.2.3 PPI++.

5 Experiments

Protocol. Each repetition draws n labelled and N unlabelled records i.i.d. *with replacement* from a pool of the `ppi-python` 0.2.3 data (Table 1); the pool's empirical distribution is the population, so the truth is the pool mean (for OLS, the pool fit), Y is hidden on the unlabelled draw, and every method sees the same draw (paired comparisons). Unless stated, $N = \min(\text{pool size}, 10^4)$ (the validity grid uses 2×10^4). Coverage is reported with Wilson 95% intervals and widths as paired ratios with confidence intervals; seeds, versions, commands and expected runtimes are in the reproduction notes. Pools without a population truth or with a shift are used only as stated in the text where they appear.

5.1 Validity of the baselines (C0)

Width is only meaningful at valid coverage, so we first measure it. Figure 1 gives the coverage of nominal 95% intervals for classical, PPI and PPI++ (Wald, t, studentised bootstrap) on seven pools, 2000 repetitions per cell, $n \in \{15, 25, 50, 100, 200, 500\}$. Wald PPI++ undercovers at small n on skewed or rare-outcome pools: on `census_income` its coverage is 0.755 at $n = 15$, 0.907 at $n = 200$ and 0.926 at $n = 500$, and the Monte-Carlo standard error is about 0.5 percentage points at 2000 repetitions. The t quantile barely helps (0.787, 0.909, 0.928). The studentised bootstrap repairs most of this on `census_income` (0.906, 0.929, 0.951) but over-corrects at $n = 15$ on the small, rare-outcome pools (`forest` 0.658, `alphafold` 0.698). On the other six pools the shortfall of PPI++ Wald is 1.1 to 7.9 percentage points at $n = 50$ and 0 to 2.2 at $n = 200$, and at $n = 500$ it is within 1 percentage point of nominal on all six (0.942 to 0.953); `gene_expression` is the closest throughout (0.946 at $n = 100$). The consequence for what follows is that the useful comparison of widths is *like for like*: an adjuster is compared with PPI++ calibrated in the same way, and no comparison against an undercovering interval is read as a saving.

5.2 Population headroom of control variates built from (f, X) (C1)

How much shorter can any adjuster be, if estimated well? Figure 2 shows the asymptotic width ratio to PPI++ (large N) of each adjuster class fitted on the whole pool with 2-fold cross-fitting, with the standard deviation over ten random splits. On the seven pools the best ratio is 0.988 (`gene_expression`, isotonic), 0.995 (`census_income`, gradient boosting on (f, X)), 0.999 or above elsewhere: at most about 1.2% in width for the classes tried (a lower bound on the true headroom, since the adjusters are fitted on half-pools), and the best ratio lies inside the grey band of $\pm 1\%$ for six of the seven pools. PPI with $\lambda = 1$ is wider than PPI++ on `gene_expression` (ratio

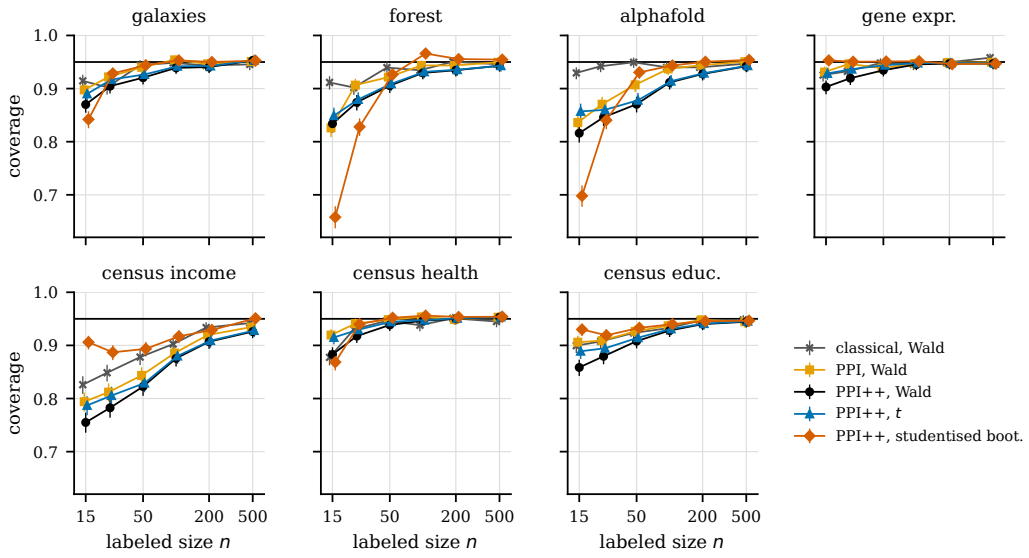


Figure 1: Coverage of 95% intervals against labelled size n on seven pools (2000 repetitions per cell; i.i.d. with replacement from the pool; $N = \min(\text{pool}, 2 \times 10^4)$; truth = pool mean). Bars: Wilson 95% intervals; line: nominal level.

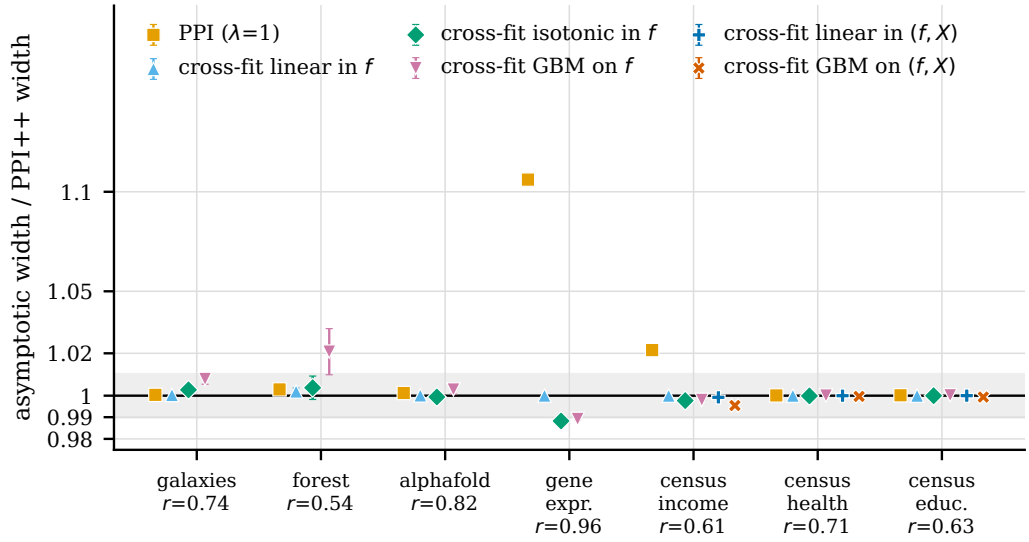


Figure 2: Population headroom: asymptotic CI-width ratio of the best control variate in each class to power-tuned PPI++ (1 = parity, < 1 narrower). Each class is fitted by 2-fold cross-fitting on the full pool; error bars are the standard deviation over ten random splits (split noise only). Grey band: $\pm 1\%$. $r = \text{corr}(Y, f)$.

1.106) and `census_income` (1.022), where f is on a different scale from Y; classical intervals are 1.19 to 3.73 times wider than PPI++. Ratios above one for flexible classes (e.g. boosting on `forest`, 1.021) reflect the estimation error of an adjuster fitted on half of a 1596-record pool, not a loss of the bound. For the mean on these pools PPI++ is therefore within about one percent of what any adjuster of the classes tried could reach; whether even that is attainable at the sizes of a labelled set is the question of the following sections.

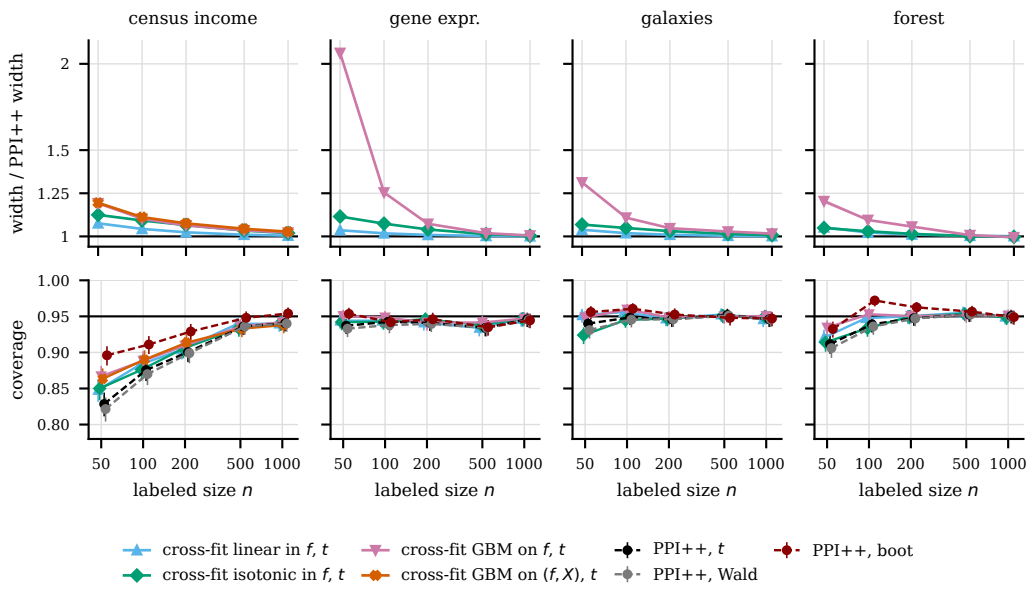


Figure 3: Mean estimation on four real pools. Top: paired width ratio of each cross-fitted adjuster to PPI++ calibrated the same way (t quantile; < 1 is narrower), geometric mean with 95% band, log axis; the cubic and the Wald and bootstrap variants are in the delivered summaries. Bottom: coverage of the 95% interval with Wilson bars; PPI++ (Wald, t, bootstrap) dashed. $n = 50$ to 1000, 2000 repetitions per cell, i.i.d. with replacement from the pool, $N = \min(\text{pool}, 10^4)$, truth the pool mean.

5.3 Finite-sample width and coverage on real pools (C2)

We now fit the adjusters of Sec. 4 from the labelled draw only and compare each interval with PPI++ *calibrated the same way* (Wald with Wald, t with t, bootstrap with bootstrap), paired on the same draws. The grid covers the four pools of the main text (`census_income`, `gene_expression`, `galaxies`, `forest`), $n \in \{50, 100, 200, 500, 1000\}$ and 2000 repetitions per cell, with $N = \min(\text{pool}, 10^4)$; the three further pools are in the appendix. Over all 270 cells (pool, n, adjuster, calibration), 252 are significantly *wider* than the like-for-like PPI++ (the confidence interval of the paired width ratio lies above 1), 9 are indistinguishable from parity and 9 are significantly narrower. All nine narrower cells are at $n = 1000$ on `forest` (gradient boosting 0.995, isotonic 0.997) and `gene_expression` (cubic polynomial 0.998); the largest saving is 0.55%. The cells are not independent: the three calibrations use the same draws, so the nine are three (pool, adjuster) combinations. So *on these real pools and for the mean, no adjuster shortens the interval in any practically relevant way*.

The loss is the estimation cost anticipated in Sec. 3 and falls with n. At $n = 50$ the like-for-like ratios of the affine-in-f adjuster, which is PPI++ up to cross-fitting, are 1.04 to 1.08; for isotonic recalibration 1.05 to 1.13, for gradient boosting on f 1.20 to 2.06 (`gene_expression`). They shrink with n but stay at or above 1.00 for all but the nine cells above (at most 1.03 at $n = 1000$). On `census_income`, the pool with features, adjusters that use X do not help: at $n = 200$, linear in (f, X) is 1.027 and boosting on (f, X) 1.076, against a population headroom of 0.5% (Sec. 5.2), which is smaller than the estimation cost, and not recovered within $n \leq 1000$ (1.005 and 1.027 there). Using studentised bootstrap in place of Wald for both sides gives the same pattern (82 of 90 bootstrap cells significantly wider, 3 narrower). The same holds as the labelled-to-unlabelled ratio varies: on `census_income` at $n = 200$ with $N = 200$, 2000 and 20000, boosting on (f, X) is 1.029, 1.068 and 1.078 times as wide as PPI++, linear in (f, X) 1.010, 1.024 and 1.027; a larger unlabelled set makes the affine tuning harder to beat, in line with the factor $1/(1 + \rho)$ in (3).

Coverage. Coverage does not rescue the adjusters: where PPI++ undercovers, the adjusters undercover too, usually slightly less. On `census_income` at $n = 200$ Wald coverage is 0.899 for PPI++ and 0.902 to 0.912 for the adjusters; at $n = 1000$ 0.940 for PPI++ and 0.938 to 0.943 for the adjusters.

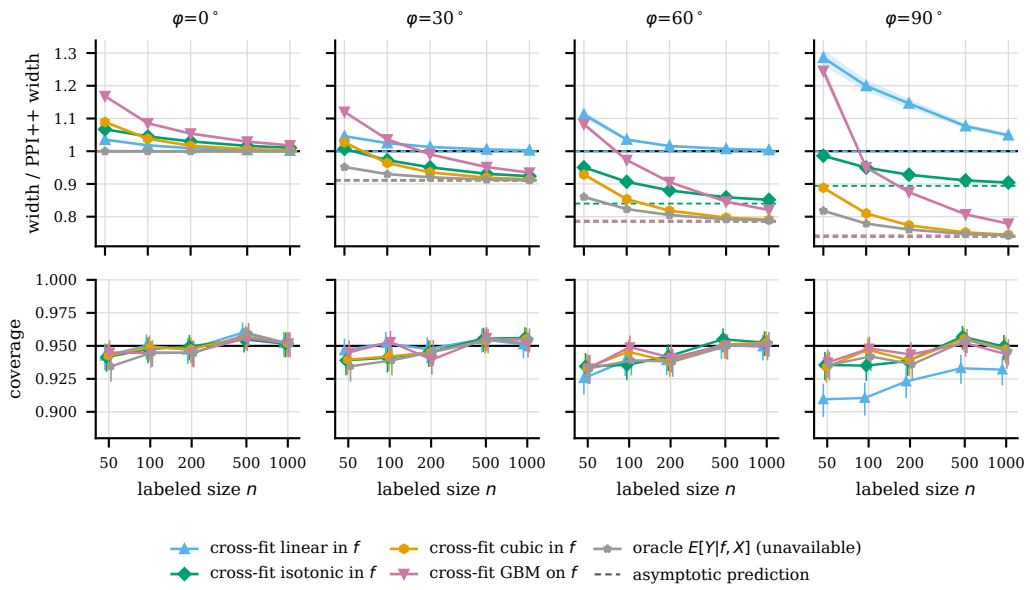


Figure 4: Knob family at $\rho = 0.1$ ($N = 10n$), 2000 repetitions per cell, known truth. Top: paired width ratio to Wald PPI++ (solid, 95% band); dashed horizontal lines: asymptotic prediction for the best adjuster of the class; grey: oracle $E[Y|f, X]$. Bottom: coverage with Wilson bars. The headroom appears only above a threshold n that grows with the flexibility of the adjuster. Other scenarios and $\rho = 1$: Appendix A.

Calibrating by the bootstrap lifts PPI++ itself to 0.929 at $n = 200$ and 0.954 at $n = 1000$, and that calibration is not free: the bootstrap interval for PPI++ is 37% wider than its Wald interval at $n = 50$ and 15% wider at $n = 200$ on `census_income`. The like-for-like comparison is the honest one, since the Wald baseline is narrow partly because it undercovers. Classical (labelled-only) intervals are 1.1 to 1.5 times wider than PPI++ on `forest`, `galaxies` and `census_income` at all sizes, and 2.5 to 3.4 times on `gene_expression`, so the large gain in this setting is from PPI++'s use of the predictor, not from anything added on top of it.

5.4 Where headroom exists: synthetic populations with known truth (C3)

The real pools leave almost no headroom for the mean, so we ask where shortening is possible at all in populations whose truth and best control variate are known. $X \sim N(0, I_5)$, $\mu(X) = X_1 + \frac{1}{2}X_2$ and $f = \mu(X)$ unless stated. The *knob family* sets $E[Y|f] = \cos\phi\, z + \sin\phi\,(z^2 - 1)/\sqrt{2}$ with z the standardised f: $\phi = 0°$ is a calibrated predictor (no headroom) and $\phi = 90°$ a purely quadratic dependence that no affine map of f can use. The named scenarios are `calibrated`; `heavy_tail` (calibrated, t_3 noise); `monotone_miscal` ($E[Y|f] = \text{logit}\, f$ up to scale); `nonmonotone` ($E[Y|f] = f^2 - \text{const}$); `X_informative` ($Y$ depends on a feature f misses); `subgroup_bias` (f biased by $+1.5$ on a subgroup); `f_noise_X_informative` (f is pure noise, X carries the signal); and `pure_noise` (neither carries signal). We use $n \in \{50, 100, 200, 500, 1000\}$ (knobs) or $\{50, 200, 500\}$ (named), $\rho \in \{1, 0.1\}$, and 2000 (knobs) or 1000 (named) repetitions. The `oracle` control variate uses the true $E[Y|f, X]$; it is not implementable, and is the bound for adjusters of the classes of Sec. 4. Ratios are paired width ratios against Wald PPI++ (Wald against Wald).

Theory. The asymptotic ratio of Sec. 3 predicts the oracle's measured ratio closely once $n \geq 200$: in the 56 cells with $n \geq 200$ the measured oracle ratio exceeds the prediction by a median of 0.05% and at most 6.6% (`nonmonotone`, $\rho = 0.1$, $n = 200$), and in the 32 cells with $n \leq 100$ by at most 22.5% (`nonmonotone`, $n = 50$). For the knobs $\phi = 30°, 60°, 90°$ at $\rho = 0.1$, $n = 1000$ the oracle is 0.912, 0.788 and 0.743 times as wide as PPI++, against predicted 0.910, 0.784 and 0.738; at $\rho = 1$ it is 0.961, 0.894 and 0.866 against 0.961, 0.894 and 0.866. The attainable saving is thus a property of the population and of ρ: up to 26% at $\rho = 0.1$ and 13% at $\rho = 1$ across the knobs.

What is realised. The adjuster must match the shape of the headroom and then pay for learning it. In the knob family at $\rho = 0.1$ the cubic in f is the first to cross below parity: its ratio at $n = 50, 100, 200, 500, 1000$ is $0.929, 0.853, 0.818, 0.797, 0.791$ for $\phi = 60°$ and $0.888, 0.810, 0.774, 0.752, 0.745$ for $\phi = 90°$ (oracle 0.788 and 0.743), and 1.027, 0.964, 0.935, 0.919, 0.914 for $\phi = 30°$, significantly below 1 from $n = 100$. Gradient boosting on f needs $n \geq 100$ to 200 (0.779 at $\phi = 90°$, $n = 1000$); isotonic regression gains at small n but stalls (0.904 at $\phi = 90°$, $n = 1000$), since it cannot represent a non-monotone map. The affine cross-fit, the PPI++ ablation, never gains at $\rho = 0.1$ and where the best affine slope is nearly zero it is clearly wider than PPI++ (1.286 at $\phi = 90°$, $n = 50$, 1.049 at $n = 1000$); at $\rho = 1$ it stays within 3% of parity. At $\phi = 0°$ every adjuster is wider than PPI++ at every n (isotonic 1.067, cubic 1.089, boosting 1.168 at $n = 50$, $\rho = 0.1$; 1.010, 1.003 and 1.018 at $n = 1000$). The named scenarios agree. `nonmonotone`, $\rho = 0.1$, $n = 500$: cubic 0.576 (oracle 0.573), boosting on f 0.668, isotonic 0.855, affine 1.092. `calibrated` and `heavy_tail` have no headroom and no cross-fitted adjuster goes below 1.001 at any n, and in `pure_noise` the affine cross-fit is 10.5% and 5.8% wider than PPI++ at $\rho = 0.1$, $n = 200$ and 500 while boosting is at parity (0.999); `monotone_miscal` has little (oracle 0.976, 0.963, 0.961 at $n = 50, 200, 500$, $\rho = 0.1$), of which the isotonic, cubic and (f, X)-linear adjusters reach only a part (0.994, 0.989 and 0.984 at $n = 200$).

Features. When X carries signal the linear adjuster on (f, X), with $d + 2$ parameters, is the practical route. It is significantly narrower than PPI++ already at $n = 50$ in `X_informative` (0.886 at $\rho = 0.1$, 0.947 at $\rho = 1$; oracle 0.828, 0.926), `subgroup_bias` (0.950, 0.974; oracle 0.880, 0.946) and `f_noise_X_informative` (0.777, 0.878; oracle 0.720, 0.853), and at $n = 500$, $\rho = 0.1$ is within one percentage point of the oracle in all three (0.823, 0.880, 0.711). Boosting on (f, X) is significantly narrower from $n = 200$ in the first two (0.909 at $n = 200$ and 0.864 at $n = 500$ for `X_informative`, $\rho = 0.1$) but 2% to 6% wider at $n = 50$. In `f_noise_X_informative` PPI++ is fed a predictor with no information, and looking at X shortens the interval by 22% to 29% at $\rho = 0.1$ ($n = 50$ to 500): the fixed predictor is what limits PPI++ there, not the estimator.

Coverage. For the 36 cells with $n \geq 200$ in which `lin_fX` or `poly_f` is at least 2% narrower (significantly), Wald coverage is 0.935 to 0.958, between -1.5 and $+1.1$ points of PPI++ in the same cell, and only four cells (all $n = 200$, cubic) have a Wilson interval that excludes 0.95, all from below (lowest 0.935). Across all cross-fitted Wald cells coverage is 0.876 to 0.957 at $n = 50$ and 0.932 to 0.956 at $n = 1000$, against 0.917 to 0.954 and 0.944 to 0.952 for PPI++. The t quantile leaves the ratios unchanged, and the studentised bootstrap moves them by 0.6 points on average (at most 2.9) for the cubic.

5.5 Regression coefficients on `census_income` (C4)

The secondary estimand is the OLS coefficient vector of income on the two covariate columns supplied with the pool (no intercept); truth is the pool fit, $N = 10^4$, 2000 repetitions per $n \in \{50, 100, 200, 500, 1000\}$. `cv_f` is the control variate built from f alone (PPI++ up to cross-fitting); `cv_cf_lin_fX` and `cv_cf_gbm_fX` use (f, X); the reference is `ppi-python`'s `ppi_ols_ci`, which tunes λ per coordinate. Ratios are against that reference (Table 2); where the two coordinates (age, sex) differ, both are given. `cv_f` matches the library within 0.5% for $n \geq 100$ (1.2% wider at $n = 50$). The linear adjuster on (f, X) is 6.2%, 7.4% *wider* at $n = 50$, 2.1%, 2.4% wider at $n = 100$, 0.7%, 0.6% narrower at $n = 200$, and 2.4% (both coordinates) narrower at $n = 500$ and 2.8% at $n = 1000$; the paired CIs exclude 1 at every n. Boosting on (f, X) is wider through $n = 500$ (18.3%, 17.4% at $n = 50$; 8.1%, 7.5% at $n = 100$; 0.7%, 0.4% at $n = 500$) and only breaks even at $n = 1000$ (0.995, 0.992). The labelled-only interval is 13% to 27% wider than PPI++ throughout. Coverage tracks the library's: at $n = 1000$ 0.930 and 0.947 for `cv_cf_lin_fX` against 0.934 and 0.949 for `ppi++_lib`; at $n = 50$ and 100 both undercover the age coefficient (0.818 and 0.851 against 0.797 and 0.844).

On this real regression problem the shortening exists, is below 3%, and needs $n \geq 200$ to 500 to appear (it is a loss of up to 7% below that); it is the real-data counterpart of the `X_informative` results of the previous subsection, at a much smaller effect.

6 Discussion and limitations

What the evidence says. With inputs fixed, the room to shorten PPI++ is set by how far $E[Y \mid f, X]$ is from affine in f, which can be estimated before choosing a method. On the seven real pools it is

Table 2: OLS of income on (age, sex) on `census_income`: paired width ratio to `ppi-python` PPI++ (< 1 narrower) and coverage of 95% intervals (Monte-Carlo SE about 0.005); 2000 repetitions, $N = 10^4$.

		width / PPI++ width					coverage			
n	coef.	classical	PPI	cv. f	lin. (f, X)	GBM (f, X)	PPI++	classical	lin. (f, X)	GBM (f, X)
50	age	1.268	1.002	1.007	1.062	1.183	0.797	0.840	0.819	0.849
	sex	1.260	1.004	1.010	1.074	1.173	0.916	0.932	0.925	0.929
100	age	1.232	1.001	1.002	1.021	1.081	0.845	0.880	0.851	0.865
	sex	1.220	1.002	1.002	1.024	1.075	0.929	0.938	0.934	0.930
200	age	1.194	1.001	1.000	0.993	1.037	0.897	0.918	0.896	0.900
	sex	1.181	1.002	1.000	0.994	1.032	0.939	0.948	0.938	0.938
500	age	1.163	1.001	0.999	0.976	1.007	0.913	0.918	0.918	0.917
	sex	1.151	1.002	0.999	0.976	1.004	0.942	0.934	0.942	0.941
1000	age	1.145	1.001	0.999	0.972	0.995	0.934	0.938	0.930	0.931
	sex	1.134	1.002	1.000	0.972	0.992	0.949	0.943	0.947	0.949

at most about 1.2% for the mean for the adjuster classes tried (a lower bound on the true gap), and all but nine of 270 adjuster cells are wider than PPI++ (Sec. 5.3), so a practitioner with these data should keep PPI++ and, if intervals undercover at small n, calibrate them (Sec. 5.1) rather than add an adjuster. The cases where an adjuster does pay off share two features: a large gap $\eta^2 - R^2_{\text{aff}}$, and n beyond a threshold set by the adjuster's size ($n = 50$ already for a $(d+2)$-parameter linear map, a few hundred for boosting). A cheap diagnostic suggested by the population calculation of Sec. 5.2, which we did not evaluate as a decision rule: fit the candidate adjusters by cross-fitting on the labelled set and compare their out-of-fold R^2 with that of the affine fit; if the gain in $R^2/(1+\rho)$ is below the estimation cost, expect a loss. Gains also fade as ρ grows (Sec. 3): a large unlabelled set makes PPI++ harder to beat (Sec. 5.3).

Limitations. *Sampling protocol.* Real-pool experiments resample the pool i.i.d. with replacement and take the pool mean (or fit) as truth; this is a clean finite population and not a fresh draw from a super-population, and $N \leq 2 \times 10^4$ throughout, so small ρ is covered only by the synthetic study and by the sweep at $n = 200$ (Sec. 5.3). *Assumptions.* Labels are missing completely at random, labelled and unlabelled records are exchangeable; under the $2013 \to 2014$ shift on `plankton` (Appendix A) PPI++ and the two adjusters tested lose coverage and nothing in this paper repairs it. *Scope.* Five adjuster classes with default hyper-parameters, $K = 5$ folds, mean and OLS only (no quantiles or logistic regression), a binary predictor only in the diagnostics; a better-tuned or richer learner could shift the break-even n but not the floor. *Theory.* The efficiency floor is a restated, known bound; the estimation-cost argument is first-order and we do not prove a break-even n nor test it quantitatively. *Synthetic scenarios* have headroom by construction and show what is possible, not what is typical. *Validity.* Wald, t and the bootstrap undercover at small n on skewed or rare-outcome pools (down to 0.755 at $n = 15$), and ratios are only read where coverage is comparable; we claim no finite-sample guarantee for any of the intervals. *Statistical power.* Ratios carry paired confidence intervals; a ratio within 0.5% of 1 is called parity or a negligible effect, not a finding. Intervals that are shorter with no loss of coverage are a benefit; we see no harm beyond the stated misuse of narrower-but-undercovering intervals. The census pools are public, anonymised microdata; age and sex enter only as regression covariates, and we infer or release nothing about individuals.

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A Diagnostics and further results

Proofs of Propositions 1 and 2. Assumptions are those of Sec. 3: i.i.d. labelled and unlabelled records, $\mathrm{Var}(Y) < \infty$, $\mathrm{Var}(g(W)) \in (0, \infty)$. *Variance.* The labelled and unlabelled records are independent, so $\mathrm{Var}(\hat{\theta}_{g,\lambda}) = \mathrm{Var}(Y - \lambda g)/n + \lambda^2 \mathrm{Var}(g)/N = \frac{1}{n}\{\mathrm{Var}(Y) - 2\lambda \mathrm{Cov}(Y, g) + \lambda^2(1+\rho)\mathrm{Var}(g)\}$. This is a convex quadratic in λ, minimised at $\lambda^\star = \mathrm{Cov}(Y, g)/\{(1+\rho)\mathrm{Var}(g)\}$, where it equals $\frac{\mathrm{Var}(Y)}{n}\{1 - R_g^2/(1+\rho)\}$ (Eq. 3). *Proposition 1.* A Wald interval has half-width proportional to the estimated standard deviation, which is consistent for the square root of the variance above, so by Slutsky's theorem the ratio of two widths converges to the ratio of the square roots of the minimal variances; with Eq. (3) for g_1, g_2 the common factor $\mathrm{Var}(Y)/n$ and the quantile cancel. *Proposition 2.* Write $m = \mathbb{E}[Y \mid W]$ and $V = \mathrm{Var}(m)$, so $\mathrm{Var}(Y) = \mathbb{E}\,\mathrm{Var}(Y \mid W) + V$ (the term $\mathrm{Var}(Y \mid W)$ in the statement is its expectation). Since g is a function of W, $\mathrm{Cov}(Y, g) = \mathrm{Cov}(m, g)$, hence $R_g^2 = \mathrm{corr}^2(m, g)\,\eta^2 \le \eta^2$ by Cauchy–Schwarz, with equality iff g is almost surely an affine function of m. Substituting η^2 in Eq. (3): $\mathrm{Var}^\star(g) \ge \frac{1}{n}\{\mathrm{Var}(Y) - \frac{V}{1+\rho}\} = \frac{1}{n}\{\mathbb{E}\,\mathrm{Var}(Y \mid W) + \frac{\rho}{1+\rho}V\}$, and $\frac{\rho}{n(1+\rho)} = \frac{1}{n+N}$ gives the second form. □We do not reprove the statement that the bound is the semiparametric efficiency bound; it is cited as known [Angelopoulos et al., 2023b].

Population headroom per pool and class. Table 3 gives the numbers behind Fig. 2 (asymptotic width ratio to PPI++, mean ± SD over ten 2-fold splits).

Table 3: Asymptotic CI-width ratio to power-tuned PPI++ ($N \to \infty$; 1 = parity, < 1 narrower) per pool and control-variate class; mean ± SD over ten random 2-fold cross-fitting splits. "–": no features X in the pool.

pool	PPI	lin. f	isot. f	GBM f	lin. (f, X)	GBM (f, X)
galaxies	1.000±.000	1.000±.000	1.003±.001	1.008±.002	–	–
forest	1.003±.000	1.002±.002	1.004±.005	1.021±.011	–	–
alphafold	1.001±.000	1.000±.000	0.999±.001	1.003±.002	–	–
gene_expression	1.106±.000	1.000±.000	0.988±.000	0.989±.001	–	–
census_income	1.022±.000	1.000±.000	0.998±.000	0.998±.000	0.999±.000	0.995±.000
census_healthcare	1.000±.000	1.000±.000	1.000±.000	1.000±.000	1.000±.000	1.000±.000
census_education	1.000±.000	1.000±.000	1.000±.000	1.000±.000	1.000±.000	0.999±.000

Label shift (`plankton`). The predictor is binary. Without shift (labelled and unlabelled drawn from the 2013 pool, $N = 10^4$, 2000 repetitions) the like-for-like ratio of the cross-fitted affine and isotonic adjusters is 1.215 and 1.267 at $n = 25$, 1.039 at $n = 100$ and 1.003 at $n = 1000$, and PPI++ covers 0.856, 0.911 and 0.943 at those sizes: the same picture as Sec. 5.3. Under the stress test (labelled from 2013, unlabelled from 2014, truth the 2014 mean of Y), the exchangeability assumption fails: PPI++ covers 0.741 at $n = 100$, 0.607 at $n = 200$ and 0.349 at $n = 1000$, the two adjusters 0.768/0.768, 0.622/0.624 and 0.352/0.358, and the labelled-only interval 0.000 at $n = 1000$. PPI with $\lambda = 1$ covers 0.952, 0.944 and 0.946 at the same sizes; we did not isolate why, but it is consistent with power-tuning shifting weight toward the biased labelled mean. In this regime a shorter interval is not a saving because no interval tested is valid; we have no remedy.

Unlabelled records without outcomes (`ballots`). Only the 1027 labelled ballots have a truth, so two diagnostics are run. *Subsample*: n labelled drawn without replacement from the 1027 and the rest as unlabelled; truth is the 1027-ballot mean. Coverage is 0.98 to 1.00 for PPI++ at $n \geq 25$ (the finite population fixes the truth; this is not a validity test of the i.i.d. theory) and the adjusters are 0.0% to 2.2% wider. *Width only*: unlabelled are the 78,150 ballots with predictions only. The isotonic adjuster is 2.4%, 2.1%, 1.7%, 1.2% and 0.6% wider than PPI++ at $n = 25, 50, 100, 200, 400$, and the labelled-only interval is 5.0 to 11 times wider. No coverage is claimed.

Further figures. Figures 5–9 extend Figs. 3 and 4 and follow the same layout (paired like-for-like width ratio above, coverage below).

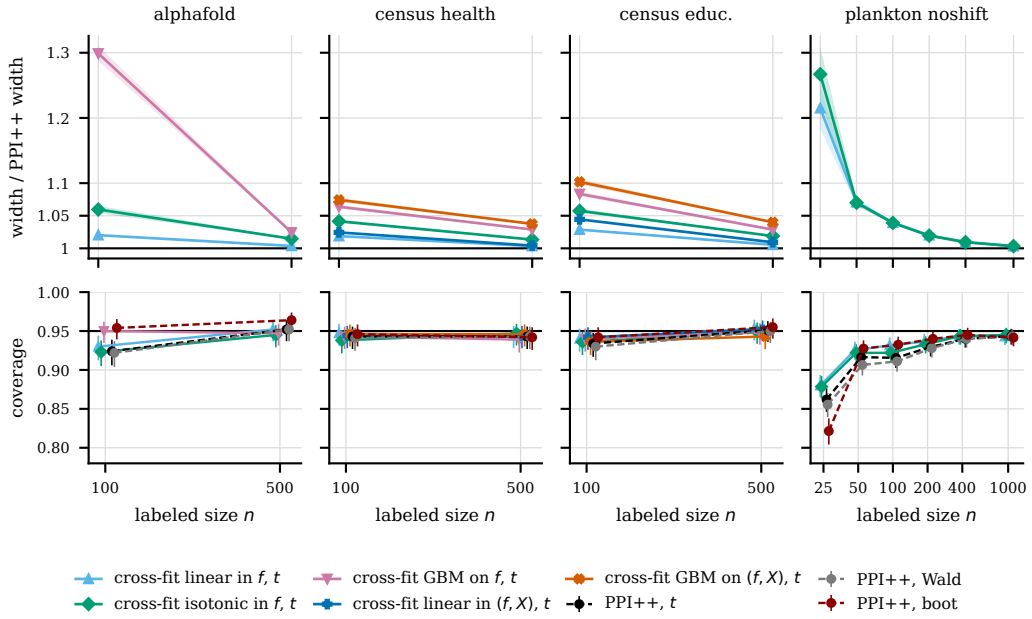


Figure 5: Mean estimation on `alphafold`, `census_healthcare`, `census_education` ($n = 100, 500$, 1000 repetitions) and on `plankton` without shift (n from 25 to 1000, 2000 repetitions).

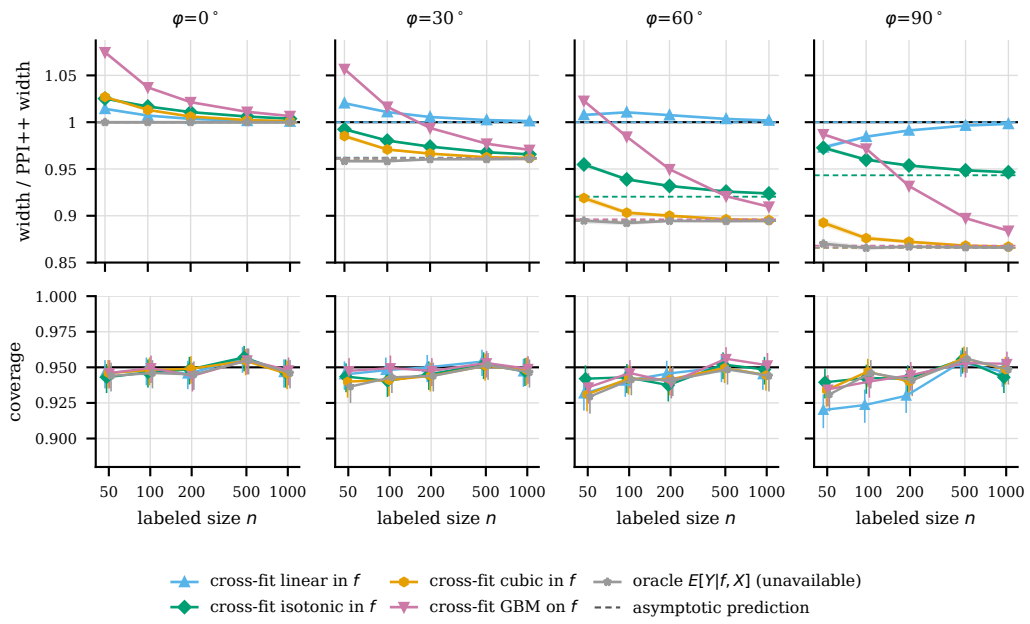


Figure 6: Knob family at $\rho = 1$ ($N = n$), 2000 repetitions per cell.

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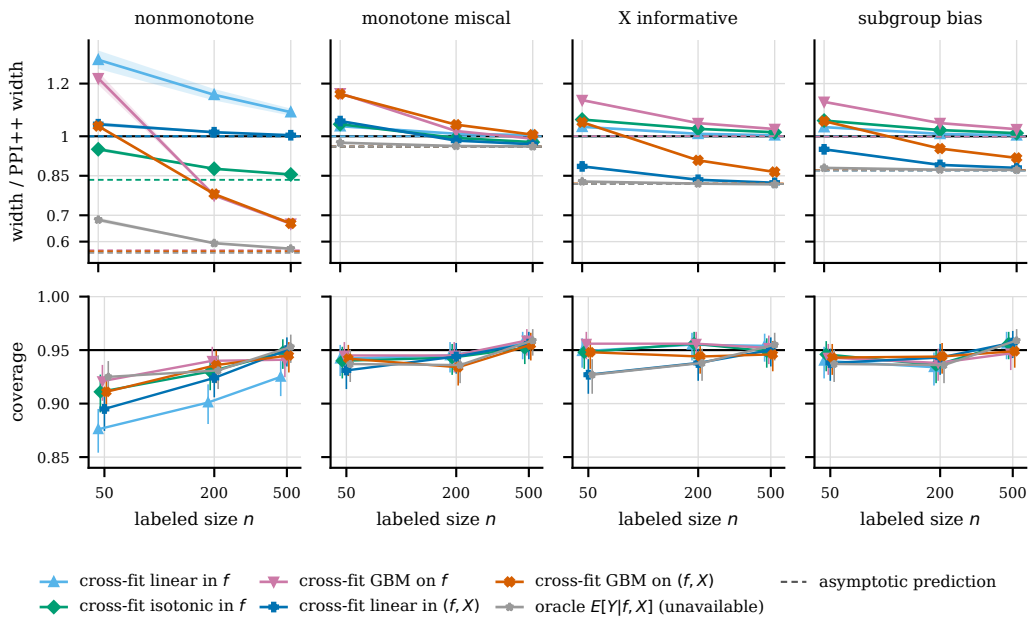


Figure 7: `nonmonotone`, `monotone_miscal`, `X_informative`, `subgroup_bias` at $\rho = 0.1$, 1000 repetitions per cell, with the (f, X) adjusters.

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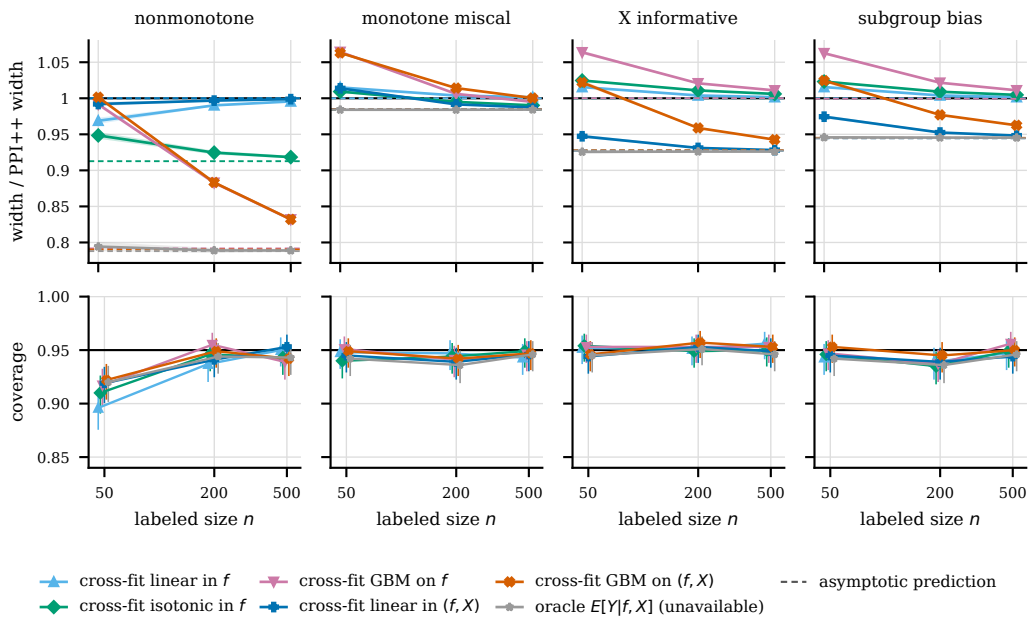


Figure 8: As Fig. 7 at $\rho = 1$.

formula; the efficiency floor is a known bound and is restated as such, and the estimation-cost argument is labelled a heuristic, not a theorem.

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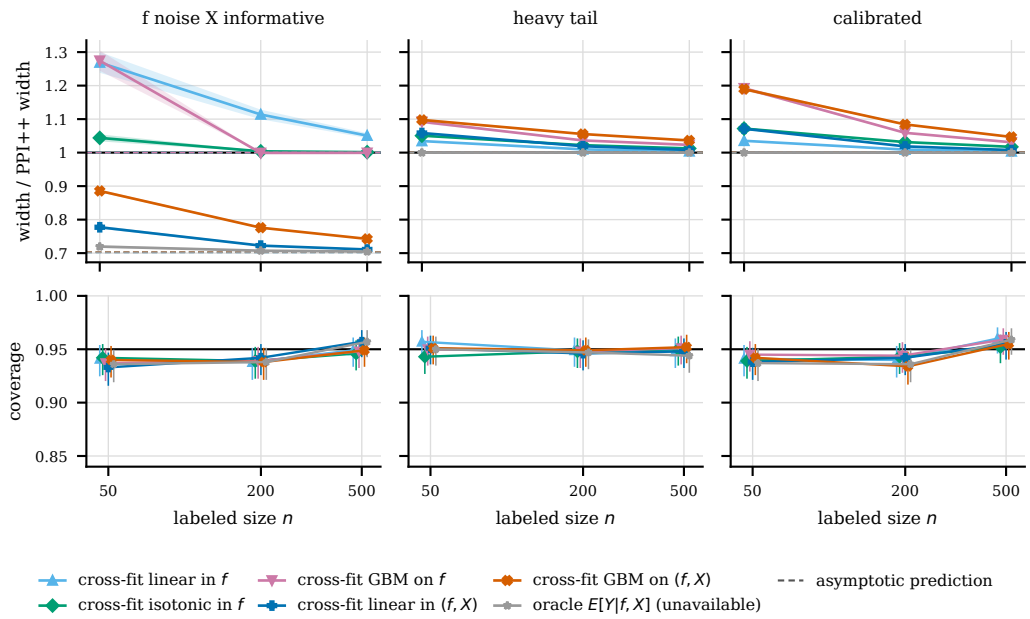


Figure 9: `f_noise_X_informative`, `heavy_tail`, `calibrated` at $\rho = 0.1$, 1000 repetitions per cell.

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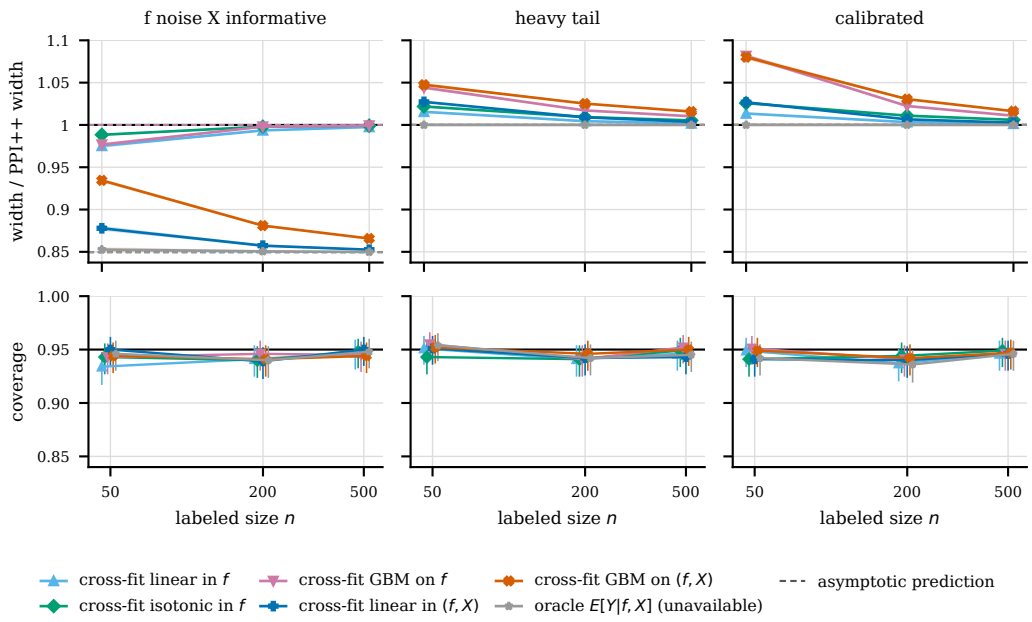


Figure 10: As Fig. 9 at $\rho = 1$.

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